RESEARCH AND EDUCATIONAL USE ONLY — NOT A MEDICAL DEVICE

Pharmacogenomic analysis summary

sample	DEMO_CONFIDENT
analysed	2026-11-06T10:18:45.884495Z
printed	2026-11-06
drugs checked	7

This is not a clinical report. It was produced by a final-year student project for research and education. It has not been clinically validated, and no qualified clinical expert has reviewed the explanatory text on this page.

Recommendation text on this page is quoted from published CPIC guidance via PharmCAT. Automated source checks are not human review: they cannot show that every statement here is supported, or that any of it is correct for the person this sample came from.

Do not change a medicine on the strength of this page. Take it to a qualified healthcare professional.

Results

Ordered by consequence, not alphabetically.

clopidogrel **Ineffective**

gene	CYP2C19
diplotype	*2/*2
phenotype	PM
evidence level	Unknown

clopidogrel: CYP2C19 *2/*2 (Poor Metabolizer) — Ineffective.

CPIC GUIDELINE — QUOTED EXACTLY

Avoid clopidogrel if possible. Use prasugrel or ticagrelor at standard dose if no contraindication.

CPIC strength of recommendation: Strong. Population: CVI ACS PCI. Implications: CYP2C19: Significantly reduced clopidogrel active metabolite formation; increased on-treatment platelet reactivity; increased risk for adverse cardiac and cerebrovascular events

— Annotation of CPIC Guideline for clopidogrel and CYP2C19 (via PharmCAT) [label rule: avoid_for_lack_of_efficacy]

azathioprine **No PGx-guided adjustment identified**

gene	TPMT
diplotype	*1/*1
phenotype	NM
evidence level	Unknown

azathioprine: TPMT *1/*1 (Normal Metabolizer) — No PGx-guided adjustment identified.

CPIC GUIDELINE — QUOTED EXACTLY

Initiate therapy with standard starting dose (e.g., 2 mg/kg/day for autoimmune diseases). During therapy, adjust doses of azathioprine based on disease-specific guidelines. It usually takes at least 2 weeks to reach steady state after each dose adjustment.

CPIC strength of recommendation: Strong. Population: general. Implications: Normal risk of thiopurine-related leukopenia, neutropenia and myelosuppression. TPMT NMs have lower erythrocyte concentrations of TGN metabolites and higher concentrations of MeMPNs compared to TPMT IMs and TPMT PMs. This is the 'normal' pattern.

— Annotation of CPIC Guideline for azathioprine and NUDT15, TPMT (via PharmCAT) [label rule: standard_dosing]

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fluorouracil **No PGx-guided adjustment identified**

gene	DPYD
diplotype	Reference/Reference
phenotype	NM
evidence level	Unknown

fluorouracil: DPYD Reference/Reference (Normal Metabolizer) — No PGx-guided adjustment identified.

CPIC GUIDELINE — QUOTED EXACTLY

Based on genotype, there is no indication to change dose or therapy. Use label-recommended dosage and administration.

CPIC strength of recommendation: Strong. Population: general. Implications: DPYD: Normal DPD activity and "normal" risk for fluoropyrimidine toxicity

— Annotation of CPIC Guideline for fluorouracil and DPYD (via PharmCAT) [label rule: standard_dosing]

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simvastatin **No PGx-guided adjustment identified**

gene	SLCO1B1
diplotype	*1/*1
phenotype	NM
evidence level	Unknown

simvastatin: SLCO1B1 *1/*1 (Normal Function) — No PGx-guided adjustment identified.

CPIC GUIDELINE — QUOTED EXACTLY

Prescribe desired starting dose and adjust doses based on disease-specific guidelines.

CPIC strength of recommendation: Strong. Population: general. Implications: SLCO1B1: Typical myopathy risk and statin exposure

— Annotation of CPIC Guideline for simvastatin and SLCO1B1 (via PharmCAT) [label rule: standard_dosing]

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codeine **Unknown**

gene	Unknown
diplotype	Unknown
phenotype	Unknown
evidence level	Unknown

codeine: no pharmacogenomic result (Unknown).

APPLICATION LIMITATION — NOT A CPIC QUOTE

CYP2D6 copy-number and structural variation is not called from the uploaded VCF in this application, so codeine guidance is not produced.

Annotation of CPIC Guideline for codeine and CYP2D6

— PharmCAT (no matching CPIC entry)

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ibuprofen **Unknown**

gene	Unknown
diplotype	Unknown
phenotype	Unknown
evidence level	Unknown

ibuprofen: no pharmacogenomic result (Unknown).

APPLICATION LIMITATION — NOT A CPIC QUOTE

This drug is outside the audited application scope; no application guidance is produced.

Annotation of CPIC Guideline for celecoxib, flurbiprofen, ibuprofen, lornoxicam and CYP2C9

— PharmCAT (no matching CPIC entry)

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warfarin **Unknown**

gene	CYP2C9
diplotype	*1/*1
phenotype	Unknown
evidence level	Unknown

warfarin: CYP2C9 *1/*1 (unknown) — Unknown.

APPLICATION LIMITATION — NOT A CPIC QUOTE

Required genomic evidence is incomplete for VKORC1. No validated automated warfarin-dose implementation is provided by this application; a gene call alone is not a complete dosing recommendation. CPIC's warfarin guidance is a dosing algorithm rather than a recommendation for each genotype: "CPIC Guideline for Pharmacogenetics-Guided Warfarin Dosing: 2017 Update [Article: 28198005] recommends that pharmacogenetic warfarin dosing be accomplished through the use of one of the pharmacogenetic dosing algorithms, as summarized in Figure 2. See CPIC guideline annotation for details." That flow chart depends in part on ancestry, which this application does not collect, so rs12777823 is not assessed. The CYP2C9, VKORC1 and CYP4F2 results are shown for reference only.

Annotation of CPIC Guideline for warfarin and CYP2C9, CYP4F2, VKORC1

— PharmCAT (no matching CPIC entry)

Unknown results (3)

An Unknown is a declined answer, not a missing one. Only the reasons about the file itself — missing positions or unusable genotypes — are something the sample provider can act on.

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codeine **Not called from an uploaded VCF**

This gene’s result depends on structural and copy-number changes that need specialised analysis of the sequencing reads, which this application does not perform. No VCF uploaded here will resolve it — it needs a specialised test or caller. The system declines to guess rather than report a genotype it cannot support.

Nothing about the uploaded file would change this result.

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ibuprofen **Result withheld**

This drug is outside the audited application scope; no application guidance is produced.

Nothing about the uploaded file would change this result.

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warfarin **Result withheld**

Required genomic evidence is incomplete for VKORC1. No validated automated warfarin-dose implementation is provided by this application; a gene call alone is not a complete dosing recommendation. CPIC's warfarin guidance is a dosing algorithm rather than a recommendation for each genotype: "CPIC Guideline for Pharmacogenetics-Guided Warfarin Dosing: 2017 Update [Article: 28198005] recommends that pharmacogenetic warfarin dosing be accomplished through the use of one of the pharmacogenetic dosing algorithms, as summarized in Figure 2. See CPIC guideline annotation for details." That flow chart depends in part on ancestry, which this application does not collect, so rs12777823 is not assessed. The CYP2C9, VKORC1 and CYP4F2 results are shown for reference only.

Nothing about the uploaded file would change this result.

Input coverage

How many of the positions each gene needs carried a usable genotype in the uploaded file. A genotype caller reads an absent position as matching the reference, so missing data can produce confident results rather than uncertain ones — which is why the count is printed whether or not it passed, and why a gene below its minimum is reported as Unknown.

gene	usable	required	minimum	sufficient
CYP2C19	35	35	100%	yes
CYP2C9	88	88	100%	yes
CYP2D6	157	157	100%	yes
CYP4F2	0	20	100%	NO
DPYD	83	83	20%	yes
NUDT15	20	20	80%	yes
SLCO1B1	35	35	100%	yes
TPMT	45	45	80%	yes
VKORC1	0	1	100%	NO

Not resolved from this file: CYP4F2, VKORC1. Any drug keyed to these genes was reported as Unknown rather than assumed normal.

Where this guidance came from

PharmCAT	3.4.0
CPIC data version	2026-07-13-11-40

CPIC table SHA-256	5010bb813ed6aaacf504c689e1f74924ddc7c58383d5e35604074d299345d6a8
allele definitions	2026-07-13-11-40
definitions SHA-256	7044c87682de18646a74b890739e2cc145ef0a3a21718a17b371dc013c7afe8f
explanations generated	24 Jul 2026
source	CPIC guidelines, retrieved via PharmCAT

Guidance reflects what CPIC published when this data was captured. CPIC revises its guidelines. A weekly repository check compares this build's CPIC data with CPIC's current release and flags differences for review; an analysis does not check at the time it runs.